

Mega evolution in MAG — a spec for the refit

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Every figure here is generated by `engine/mega_decision_census.js` into `data/mega-decision.json`. Nothing is typed. Two hand-typed store counts in `docs/GAME-DIFFERENTIAL-DESIGN.md` had already drifted apart from each other and from the file within hours of being written, and the census exists so this document cannot repeat it.

1. Where this came from

Will, 2026-08-06: "WHAT IF WE ADD MEGA EVOLUTION TO THE MAG WEIGHTS", then two proposed features — "SO IF MEGA I OUTSPEED" and "IF MEGA I GAIN WEATHER CONTROL" — and then the constraint that reorganises the whole design:

"although it needs to be a separate score than from the move selection or switching because you can mega and act at the same turn"

That constraint is correct and it is not a detail. MAG is a conditional logit over a choice set: $P(\text{pick } j) = \frac{\exp(w \cdot x_j)}{\sum \exp(w \cdot x_k)}$. A softmax assumes you pick **one**. Mega is not one of the options — **you mega AND you act**. Adding a `mega` row to the choice set would model the player as choosing mega *instead of* clicking Flare Blitz, which is not the game.

2. The three preconditions, measured

A feature fitted on human clicks needs the decision to be **observed**, **timed**, and **variable**. All three are asked by the census rather than assumed.

	raw ladder store	clean corpus
games	47,210	8,493
games with ≥1 mega	93.1%	98.53%
both sides megaed	61.32%	78.54%
mega events	74,944	15,803
share executed on turn 1	47.78%	40.47%

Observed: yes. **15,803 mega events in the clean corpus** — the number that matters, because anything about *judgement* must be fitted on games not dominated by bots. (An earlier statement in conversation put this at 74,944; that is the RAW figure and it is the wrong one to quote for a fit.)

Timed: yes, and more so once the corpus is cleaned. **59.53% of clean megas are held past turn 1**. If this were ~100% on turn 1 there would be no decision to fit and ROADMAP #31's "today it is a default — the lead keeps it" would be a harmless simplification. It is not.

Variable: yes, and the shape is narrower than it first appears — see §3.

3. What megaing actually buys, across 98 mega formes

	count of 98
gain base Speed	61

	count of 98
gain no Speed at all	26
set weather on evolving	4
conditional speed ability	1
ability not overwritten	21

Three corrections to what was said in conversation before this was measured:

- **Weather is 4 formes, not a family** — `charizardmega` → `sun`, `tyranitar` → `sand`, `abomasnowmega` → `snow`, `froslassmega` → `snow`. It matters by **usage**, not by breadth: Charizard-Mega-Y is the single most-used mega in the store.
- **Conditional speed abilities are ONE forme** — `swampertmega`, Swift Swim, ×2 in rain. It was described as a group; it is not.
- **Charizard-Mega-Y gains ZERO base Speed**, and so does Charizard-Mega-X. The most-used mega in the format is a counter-example to "*if mega I outspeed*". The largest gains are `absolmegaz` +76, `beedrillmega` +70, `lucariomegaz` +61 — none of which was the example reached for from memory.

This variation is the whole argument for a weight rather than a rule. Blastoise-Mega gains no Speed and sets no weather; Absol-Mega gains 76. "*Mega is good*" is false as a rule and estimable as a conditional weight.

4. The design, given that you mega AND act

Three options were considered. **The first is wrong for the reason in §1** and is recorded so it is not re-proposed.

✗ 4a. Add `mega` as a row in the choice set

Models mega as competing with your move. It does not. Rejected.

✓ 4b. Mega is a BOARD TRANSFORMATION; score the same moves twice

The action space is a **product**, not a union: `(mega? × move × target)`.

```
score(mega=no, m) = MAG(m)  evaluated on the CURRENT board
score(mega=yes, m) = MAG(m)  evaluated on the POST-MEGA board
```

This is why almost nothing new needs to be written. "If mega I outspeed" is not a feature — it is `movesFirst`, recomputed on a board where the Speed stat is different. "If mega I gain weather control" is not a feature — it is the existing weather-dependent features, recomputed on a board where sun is up. **The existing 58 already say these things; they have simply never been offered the post-mega board.**

That also sidesteps ROADMAP #79's admission rule for most of it: a re-scored existing feature needs no new justification, where a new column would have to earn its place on held-out likelihood.

Cost: the candidate set doubles, ~64 pairs → ~128, for the one Pokémon that can still evolve. That is a speed budget question (#76, #61), and it is the reason this is a spec rather than a patch.

✓ 4c. One genuinely new thing — the resource

Re-scoring on the transformed board captures everything about **this** turn and nothing about the **cost**. The mega is once per battle. Spending it now forecloses spending it on turn 4, and no re-scored feature can see that, because every feature MAG has is about the present board.

So the new columns are small and are about the *option*, not the *effect*:

candidate	says
megaSpent	this action consumes the side's one mega
megaUrgency	the body holding the stone is at risk of fainting before it can be spent

megaUrgency is Will's own question from 2026-08-06 — "*how many of the raichus and all these megas died before they could evolve?*" — and it is the honest form of the resource cost: the option is worth less when it may expire.

5. What this can and cannot decide

MAG can learn WHEN A HUMAN MEGAS. It cannot tell you whether megaing is CORRECT.

MAG is behaviour cloning (docs/GLOSSARY.md §3) and caps at its target by construction. A once-per- battle resource is valued by what it is worth *later*, which is a position-and-future question, not an action-on-this-board question. That is CLAUDE.md 's rule that features are per-model: MAG scores an ACTION, a value function scores a POSITION.

- **MAG + mega** → a better opponent model (*will they mega this turn?*), a better prior, better pruning. Directly useful, and currently **absent entirely** for a decision present in 98.53% of clean games.
- **Whether WE mega** → search and the value function. **ROADMAP #31's engine half landed in 3.62.2; the DECISION half — a search that chooses whether to spend the mega — has not.** Not this spec's job.

6. Preconditions before any of this is fitted

1. **~~The engine must be able to represent the decision.~~ DONE, 3.62.2.** MEDICHAM megaed at BUILD time, so there was no post-mega board to score against and no mega action to fit. ROADMAP #31 landed: evolution is a choice taken mid-turn, every offered choice is taken identically by both engines, and the differential now tests mega bodies with no stone stripped. **The post-mega board §4b needs now exists.**
2. **Fitting environment must match playing environment** (CLAUDE.md). Weights fitted with mega as a build-time fact are not valid for an engine that megas on a choice. This lands **with** the refit, not before it.
3. **ROADMAP #79's admission rule applies to §4c only.** megaSpent and megaUrgency each have to improve held-out likelihood against the set without them, or they do not ship. The §4b re-scoring is not a new feature and is not subject to it.
4. **The censoring caveat travels with any fit.** The store records a mega that HAPPENED. It cannot record one that was considered and held, nor one whose body fainted before it could be spent. The turn-share is therefore a distribution over **executed** megas and a **lower bound** on how often the decision was faced. This is stated in data/mega-decision.json 's own caveat field.

Sources

- data/mega-decision.json — every figure above, generated by engine/mega_decision_census.js
- docs/ROADMAP.md §5 — #31 (mega as a strategic decision), #79 (the admission rule), #76/#61 (speed)
- CLAUDE.md — features are per-model, facts are global; fitting environment must match playing